

Business Intelligence & Operations Analytics

Professional Case Study

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PROJECT AT A GLANCE	DETAIL
Business Domain	Trust & Safety Operations
Primary Focus	Operations Performance + Workforce Management
Analytical Capability	Workforce Demand Forecasting
Data Platform	SQL Server
BI Platform	Microsoft Power BI
Analytical Language	DAX
Development Support	Tabular Editor
Implementation Scope	MVP

Case Study Purpose

Show how a business problem was translated into requirements, data architecture, SQL reporting, Power BI analytics, validation and management decision support.

Portfolio case study • Final implementation reflects documented scope and architecture decisions.

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Implementation principle

The case study distinguishes clearly between what was implemented in the MVP and what remains a future-state capability.

1. Executive Summary

The Enterprise Trust & Safety Operations Analytics project is an end-to-end Business Intelligence solution designed to improve management visibility into operational performance, workforce capacity and future workload demand.

The solution combines Business Analysis deliverables, dimensional data warehouse design, SQL reporting views, Power BI semantic modelling and DAX-based analytics. The implemented MVP focuses on Executive Overview, Operations Performance, Workforce Management and workforce demand forecasting.

Core delivery story

Business problem → requirements → AS-IS analysis → TO-BE solution → dimensional model → SQL reporting layer → Power BI → validation → business insights.

What the project demonstrates

- Business requirements translated into measurable KPIs and dashboard requirements.
- Enterprise-style dimensional modelling using shared business dimensions.
- SQL-based reporting views separating warehouse structures from the BI reporting layer.
- Power BI dashboards designed around management and operational questions.
- Workforce capacity analysis using scheduled, logged, adherence, occupancy and staffing measures.
- Forward-looking workload analysis using Power BI native time-series forecasting.
- Explicit data validation and change management rather than unsupported assumptions.

2. Business Context & Problem

The original business requirements described a Trust & Safety reporting environment in which operational, quality, workforce, appeals and financial reporting were fragmented across departmental outputs. The AS-IS analysis identified manual extraction, Excel consolidation, KPI validation and limited cross-functional visibility as central reporting problems.

AS-IS CHARACTERISTIC	BUSINESS IMPACT
Decentralized reporting across Operations, WFM, QA and Finance	Limited enterprise-wide visibility
Manual Excel consolidation	Time-consuming and error-prone reporting cycle
Different KPI definitions / calculation approaches	Inconsistent management narratives
Separate departmental reports	Difficult cross-functional analysis
Static reporting	Limited historical exploration and drill-down
Manual KPI validation	Additional analyst effort and risk of inconsistency
Weekly reporting cycle	Reactive rather than proactive decision support

Business problem statement

Problem

Stakeholders needed a centralized, consistent analytical environment to monitor operational health, service performance and workforce capacity without relying on repeated manual consolidation.

The AS-IS process analysis documented an approximate three-hour report preparation effort, four departmental reports being consolidated, weekly reporting, high manual activity, limited historical analysis and low cross-functional visibility.

3. Business Objectives & Stakeholders

Business objectives

- Centralize operational reporting and create a consistent management view.
- Standardize KPI definitions and calculations.
- Improve visibility into SLA, quality, productivity and workforce performance.
- Support workforce capacity planning using historical demand.
- Reduce reliance on manual reporting consolidation.
- Create a scalable foundation for future advanced analytics.

Primary stakeholder groups

STAKEHOLDER	DECISION / INFORMATION NEED
Operations Leadership	Overall operational health, workload, SLA, productivity and site performance
Operations Managers	Operational trends, performance gaps and team/site comparisons
Workforce Management	Staffing, utilization, adherence, shrinkage and demand
Capacity Planning	Required versus actual workforce coverage
Quality Management	Quality outcomes and error-related analysis
Finance	Revenue and financial reporting
Business Analysts	Requirements, KPI definitions, validation and decision support

4. Requirements-to-Solution Lifecycle

The project was developed as a Business Analysis-led analytics lifecycle rather than beginning with visualization design.

STAGE	ACTIVITY	OUTPUT / PURPOSE
01	Business Problem	Identify reporting fragmentation, visibility gaps and decision-making needs.
02	Requirements	Define objectives, stakeholders, business questions, KPIs, filters and dashboard requirements.
03	AS-IS Analysis	Document current workflows, reporting handoffs, pain points and manual activities.
04	TO-BE Vision	Define a centralized, scalable reporting architecture and future-state workflow.
05	Data Model	Design facts, dimensions, relationships and shared analytical structures.
06	SQL Layer	Create warehouse structures, data generation/transformation and reporting views.
07	Power BI	Build semantic measures, dashboards, filters and contextual tooltips.
08	Validation	Validate data, relationships, KPI calculations and analytical suitability.
09	Change Management	Document scope reductions, data limitations and implementation variances.

5. AS-IS to TO-BE Transformation

AS-IS	TO-BE DIRECTION
Multiple departmental reports	Centralized analytical reporting layer
Manual Excel consolidation	SQL reporting layer feeding Power BI
KPI validation after consolidation	Standardized KPI definitions and semantic measures
Separate operational and workforce views	Integrated operational + workforce analysis
Weekly/static reporting	Interactive historical analysis
Reactive workforce planning	Historical-demand-based forward view

Final implementation architecture

BUSINESS SOURCES	DATA & REPORTING LAYER	BI / DECISION LAYER
Operations WFM QA Appeals Coaching Finance	SQL Server Dimensional Warehouse Reporting Views	Power BI Semantic Model DAX Measures Dashboards Forecasting
Business requirements	Data validation & business rules	Management insights
Shared dimensions Date • Client • Site • Employee • Process • Team • Shift	Subject-area views Operations • Workforce • QA • Appeals • Coaching • Finance	Executive • Operations • Workforce
Future integrations	Automated ETL / scheduled refresh	Future advanced analytics

Simplified representation of the final implemented analytical flow; detailed architecture artifacts are maintained in the repository.

6. Data Architecture & Final Model

The warehouse was designed around a star-schema approach with business facts surrounded by reusable dimensions. The enterprise architecture documentation uses a bus-matrix concept to map business processes to shared dimensions and provide an integrated analytical blueprint.

Implemented business processes

FACT / PROCESS	ANALYTICAL PURPOSE
Fact_Operations	Operational case activity and performance
Fact_QA	Quality audit outcomes
Fact_Workforce	Workforce activity, scheduling and capacity
Fact_Appeals	Appeal lifecycle and outcomes
Fact_Coaching	Manager coaching interventions
Fact_Finance	Revenue / financial reporting

Final implemented shared dimensions

- Dim_Client
- Dim_Process
- Dim_Team
- Dim_Site
- Dim_Shift
- Dim_Policy
- Dim_Error_Code
- Dim_Appeal_Decision
- Dim_Shrinkage
- Dim_Billing_Model
- Dim_Cost_Center
- Dim_Date
- Dim_Employee

Final-model governance

The final implemented model reflects the approved project architecture after documented implementation changes. Earlier conceptual architecture artifacts contained broader enterprise entities; the final SQL/Power BI implementation was aligned to the validated MVP model.

7. SQL Reporting & Semantic Layer

SQL Server provides the warehouse and reporting foundation. Reporting views expose reporting-ready datasets to Power BI while keeping the warehouse layer separate from the BI presentation layer.

REPORTING VIEW	PURPOSE
vw_Operations_Dashboard	Operations reporting dataset
vw_QA_Dashboard	Quality reporting dataset
vw_Appeals_Dashboard	Appeals reporting dataset
vw_Coaching_Dashboard	Coaching reporting dataset
vw_Workforce_Dashboard	Workforce and capacity reporting dataset
vw_Finance_Dashboard	Financial reporting dataset

Analytical modelling principles

- Facts represent business events or periodic operational states at a defined grain.
- Dimensions provide reusable business context for filtering and analysis.
- Surrogate keys support stable warehouse relationships.
- Conformed dimensions enable consistent cross-functional analysis.
- Reporting views simplify Power BI consumption without replacing the underlying warehouse model.

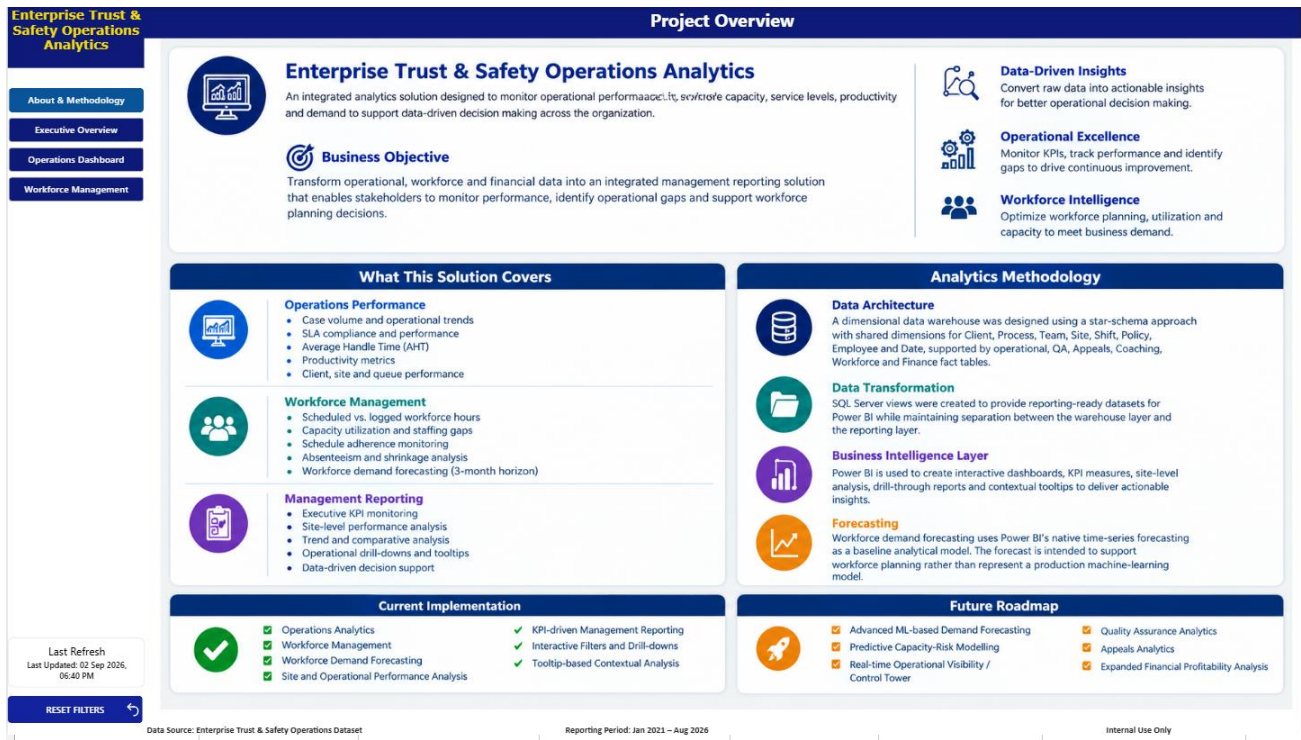
Power BI semantic layer

DAX measures were used for KPI calculations including total workload, SLA compliance, AHT, productivity, workforce hours, occupancy, adherence, staffing gaps and forecasting-support measures.

8. Power BI Solution

The final Power BI implementation prioritizes three implemented analytical experiences: Executive Overview, Operations Dashboard and Workforce Management. An About & Methodology page documents the solution approach, scope and future roadmap.

About & Methodology

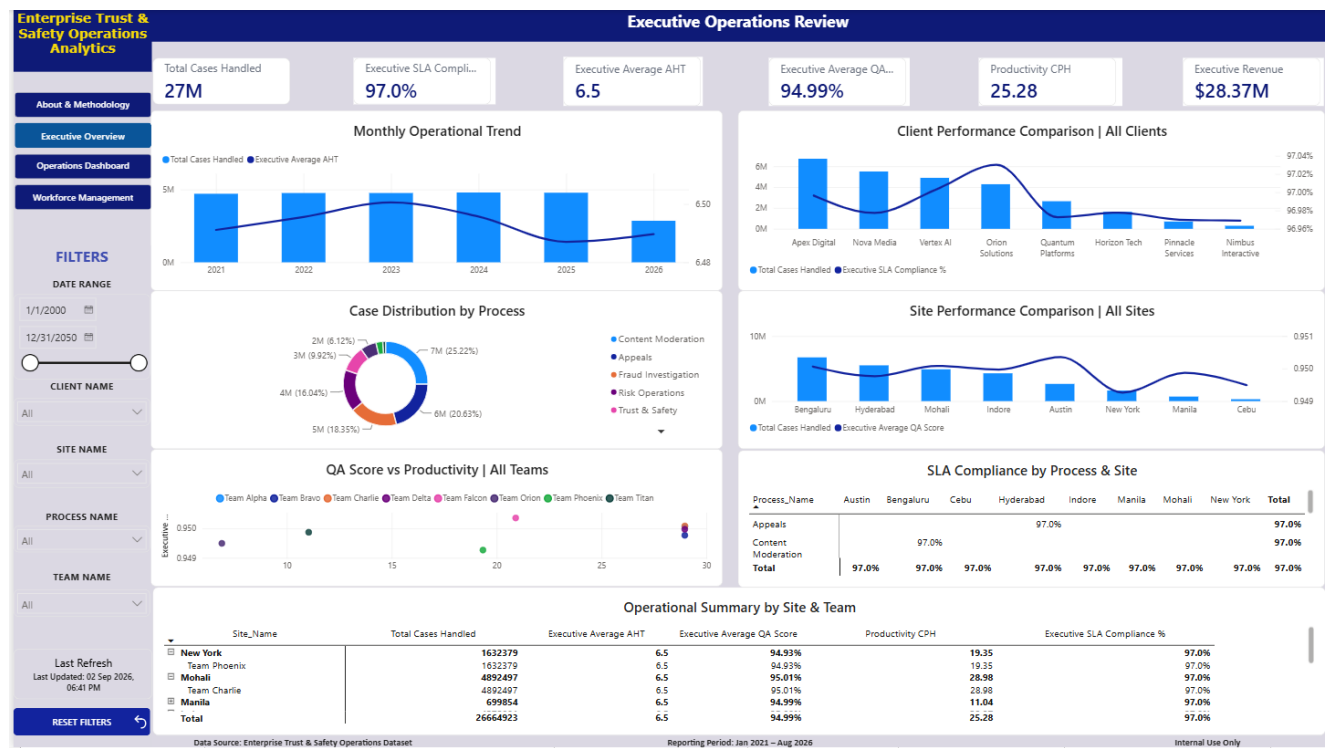


Final Power BI About & Methodology page.

The About page provides the report-level explanation of the business objective, solution coverage, methodology, current implementation and future roadmap.

Executive Overview

The Executive Overview consolidates enterprise-level operational indicators and comparative views across clients, sites and processes.

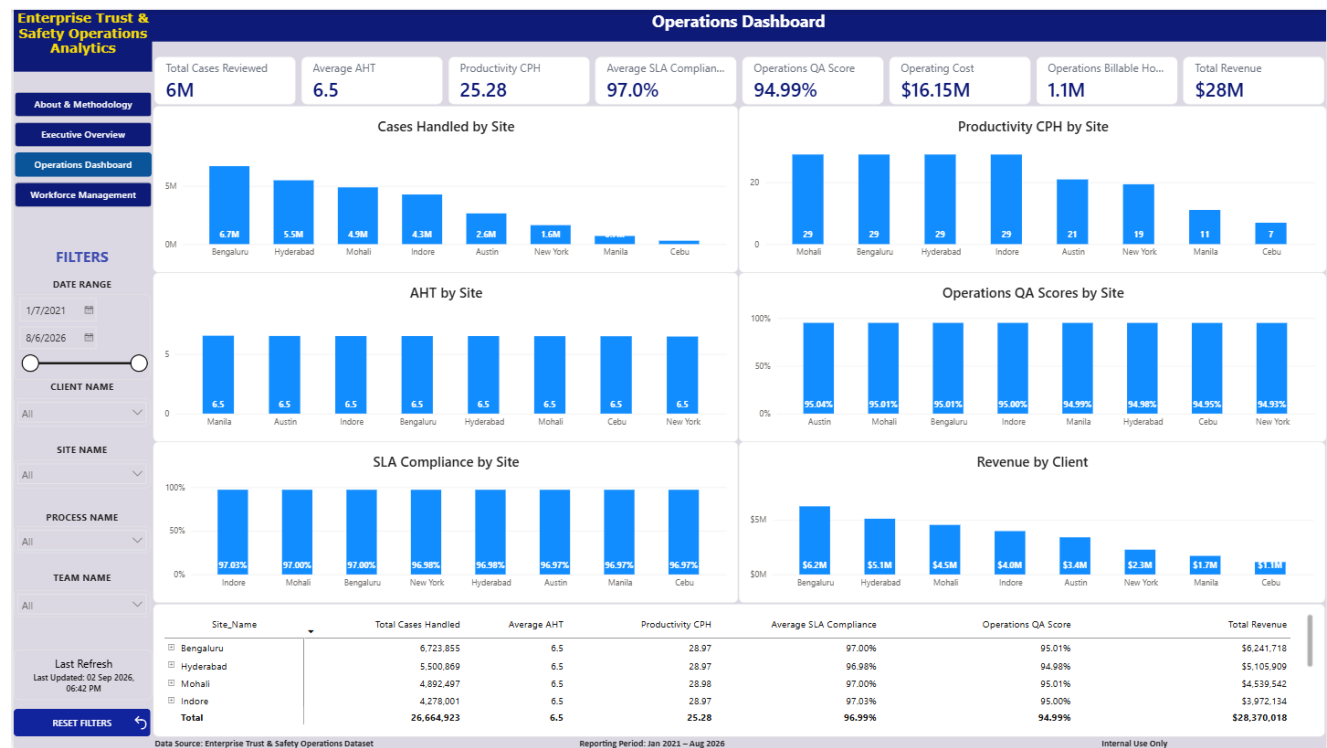


Final Power BI Executive Overview — implemented MVP.

The final page includes KPI cards for total cases handled, SLA compliance, average AHT, QA score, productivity and revenue, alongside monthly trend, client comparison, process distribution, site comparison, QA/productivity analysis and an operational summary.

Operations Dashboard

The Operations dashboard provides detailed site-level performance analysis across workload, productivity, AHT, SLA, QA and revenue.

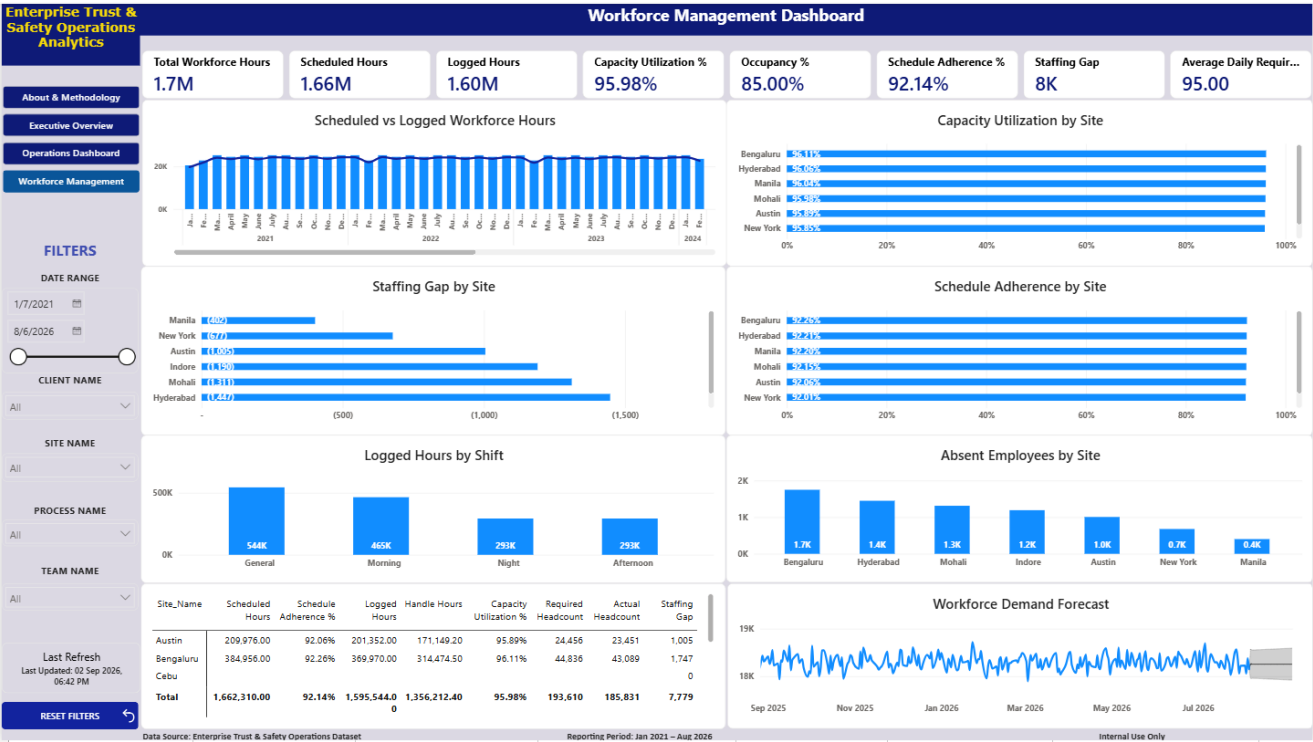


Final Power BI Operations Dashboard — implemented MVP.

The implemented page uses actual available data rather than forcing every original wireframe element into the final report. This distinction is documented through the project's change-management artifacts.

Workforce Management Dashboard

The WFM dashboard brings workforce hours, utilization, staffing coverage, adherence, absenteeism and demand forecasting together.



Final Power BI Workforce Management Dashboard — implemented MVP.

The final implementation includes Scheduled vs Logged Hours, Capacity Utilization by Site, Staffing Gap by Site, Schedule Adherence, Logged Hours by Shift, Absent Employees by Site, a site summary and the Workforce Demand Forecast.

9. Workforce Demand Forecasting

A three-month workforce demand forecast was implemented using Power BI's native time-series forecasting capability. The objective is to provide a practical baseline for forward-looking workforce planning.

What is implemented

Historical case volume → Power BI time-series model → three-month forecast → confidence interval → workforce planning context.

Analytical interpretation

- Enterprise daily demand is relatively stable across the available historical period.
- Site-level demand differs materially, making site-specific workload context more useful for capacity planning.
- The forecast is a baseline analytical capability rather than a production machine-learning model.

What this project does not claim

- No production ML forecasting pipeline was implemented.
- No model-accuracy framework using MAE, RMSE or MAPE is claimed.
- No real-time streaming/control-tower capability is implemented.

Interview-ready positioning

"I implemented Power BI native time-series forecasting as a baseline for workforce demand planning, while documenting advanced ML forecasting as a future enhancement."

10. Key Analytical Findings

01	Enterprise demand stability Historical daily case volumes show limited enterprise-level seasonality, supporting use of historical workload as a planning baseline.
02	Site-level workload concentration Average daily workload varies significantly by site, making site-level capacity planning more actionable than enterprise averages alone.
03	Workforce coverage gap Required and actual headcount differ across operating locations, providing a direct planning signal for staffing attention.
04	Data coverage matters Cebu is represented in employee master data but lacks corresponding WFM activity; the solution avoids interpreting missing activity as zero capacity.
05	Forecasting is a baseline The implemented forecast provides forward-looking demand visibility without overstating it as advanced AI/ML.

11. Data Quality & Validation

Validation was treated as part of the analytical development lifecycle. KPI reliability, relationship behaviour and source coverage were tested before promoting measures into the final dashboard.

VALIDATION AREA	APPROACH
Fact grain	Confirm that each fact table represents the intended business event / snapshot grain.
Relationships	Validate dimension-to-reporting-view relationships and filter propagation.
Date coverage	Check date ranges and ensure reporting periods are appropriate.
KPI calculations	Cross-check DAX measures against SQL/source values.
Percentage scaling	Validate whether source percentages are stored as 0–1 or 0–100 values.
Workforce coverage	Check site/employee coverage and distinguish missing records from zero activity.
Forecast suitability	Assess historical volume stability and site-level variation before using forecasting.
Visual usability	Review titles, filters, tooltips and summary tables for management readability.

Data-quality principle

Where source data was insufficient for a reliable KPI, the project documented the limitation rather than manufacturing values or presenting unsupported analysis.

12. Scope Evolution & Change Management

The original BRD defined a broader enterprise reporting initiative with six dashboards. The implementation was subsequently narrowed to a defensible MVP based on actual data availability, analytical reliability and delivery priorities.

ORIGINAL / PLANNED	FINAL IMPLEMENTATION	REASON / TREATMENT
Six-dashboard enterprise scope	Executive + Operations + WFM MVP	Scope reduced to prioritize reliable, demonstrable analytics.
Queue Utilization KPI	Not promoted	Reliable capacity field was not available for defensible calculation.
Required vs Actual Headcount visual	Workforce Demand Forecast	Forecast provides forward-looking demand visibility within available data.
Advanced AI/ML forecasting	Power BI native baseline	Advanced ML remains future roadmap.
Predictive capacity risk	Supporting measures only / not headline	Risk largely reproduced staffing-gap behaviour and was not promoted.
Finance profitability dashboard	Deferred	Available finance view did not provide reliable cost/profit fields for requested KPIs.
QA / Appeals dashboards	Deferred	Retained in architecture and documentation; not part of implemented MVP.
Real-time control tower	Future state	No streaming/real-time implementation claimed.

Why this matters

The project demonstrates controlled scope management: when a requirement could not be supported reliably by the available data, the implementation was changed or deferred and the variance was documented. This preserves analytical credibility.

13. Business Value

CAPABILITY	BUSINESS VALUE
Centralized KPI reporting	Creates a consistent management view of operational health.
Site-level analysis	Allows leaders to compare workload, productivity, SLA and quality across locations.
Workforce analytics	Provides visibility into scheduled/logged hours, adherence, occupancy and staffing gaps.
Demand forecasting	Adds a forward-looking workload signal for workforce planning.
Interactive analysis	Enables users to investigate performance by date, client, site, process, team and shift.
Validated metrics	Improves trust in reported KPIs by connecting business definitions with tested calculations.
Scalable architecture	Provides a foundation for future analytical subject areas and advanced analytics.

Business Analyst contribution

The value is not limited to dashboard construction. The project connects business questions, requirements, process analysis, data architecture, KPI governance, validation and reporting into one traceable delivery lifecycle.

14. Future Roadmap

FUTURE CAPABILITY	PURPOSE
Advanced ML-based demand forecasting	Extend baseline forecasting using richer operational and workforce drivers.
Predictive capacity-risk modelling	Move from descriptive staffing gaps toward forward-looking risk signals.
Real-time operational visibility / control tower	Support intraday monitoring and faster operational response.
Quality Assurance analytics	Operationalize the documented QA requirements and quality insights.
Appeals analytics	Analyze appeal volume, aging, overturns and resolution performance.
Expanded financial profitability analysis	Add reliable cost, margin and billing fields to support profitability analytics.

These are explicitly future-state capabilities. They are not presented as completed functionality in the current MVP.

15. Technology & Project Deliverables

TECHNOLOGY / ARTIFACT	ROLE
SQL Server / SSMS	Warehouse implementation, SQL development and validation
Power BI	Semantic model, interactive dashboards and forecasting
DAX	Business measures and analytical calculations
Tabular Editor	Bulk semantic-model and measure management
Business Requirements	Business objectives, scope, stakeholders and success criteria
Functional Requirements / User Stories	Detailed reporting and user needs
RTM / UAT	Requirements traceability and validation evidence
Data Dictionary / KPI Repository	Definitions, source mapping and business meaning
ERD / Architecture	Warehouse and enterprise analytical design
Dashboard Specifications	Business questions, KPIs, filters and visualization requirements
Change Log	Implementation variances, limitations and scope decisions

Repository evidence

The GitHub repository is organized to preserve the evidence chain from project documentation through SQL implementation, Power BI evidence and the portfolio case study.

16. Final Takeaway

The project in one sentence

An enterprise-style Business Intelligence solution that translates Trust & Safety operational and workforce requirements into a validated SQL + Power BI analytical environment for performance monitoring and workforce planning.

What this project demonstrates

- Business Analysis and requirements thinking.
- Stakeholder-oriented KPI and dashboard design.
- Dimensional data warehouse concepts.
- SQL reporting and data validation.
- Power BI and DAX development.
- Workforce capacity and demand analysis.
- Forecasting used appropriately as a baseline analytical capability.
- Scope governance and transparent change management.
- Ability to convert analytical findings into business-oriented decision support.

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